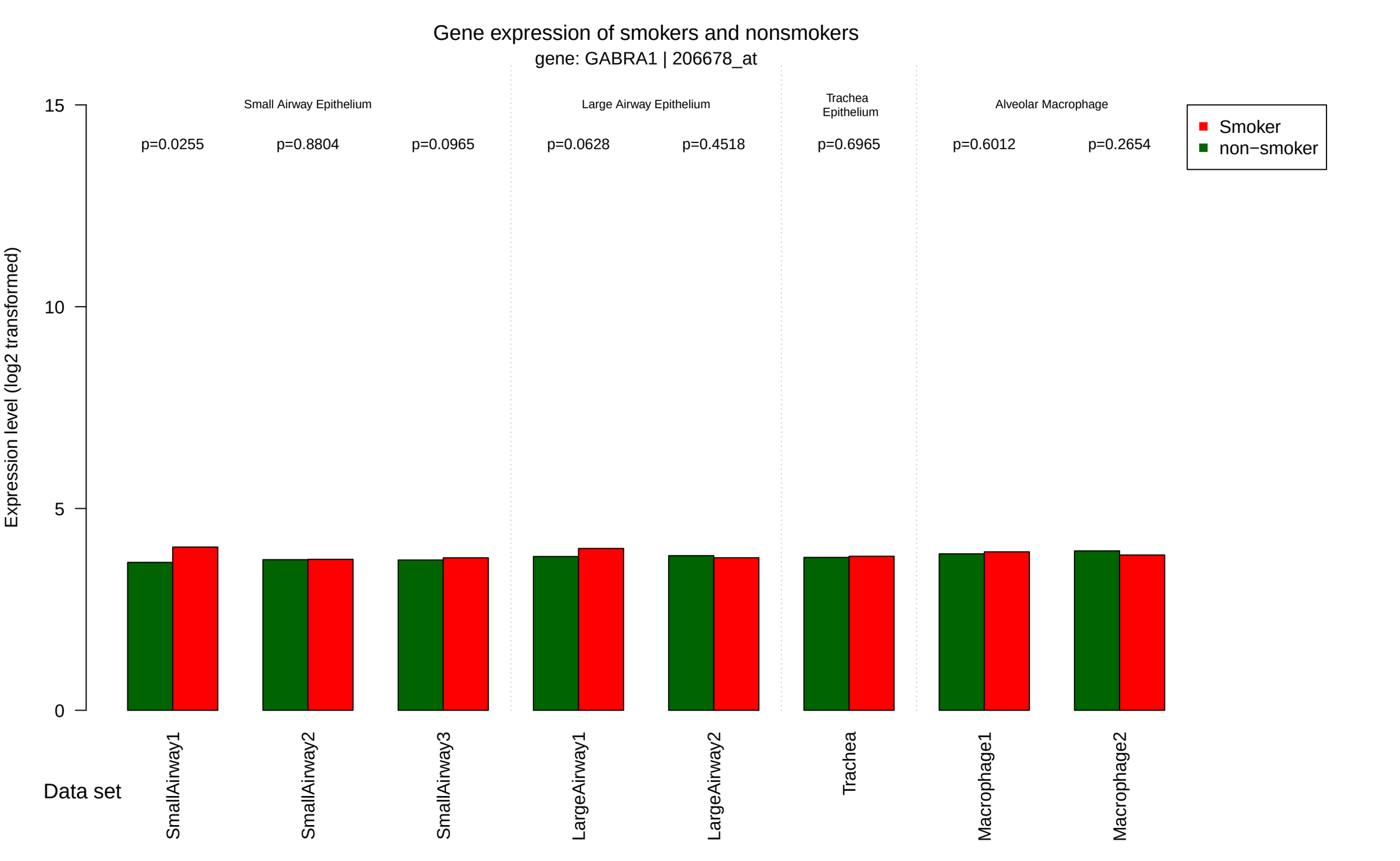
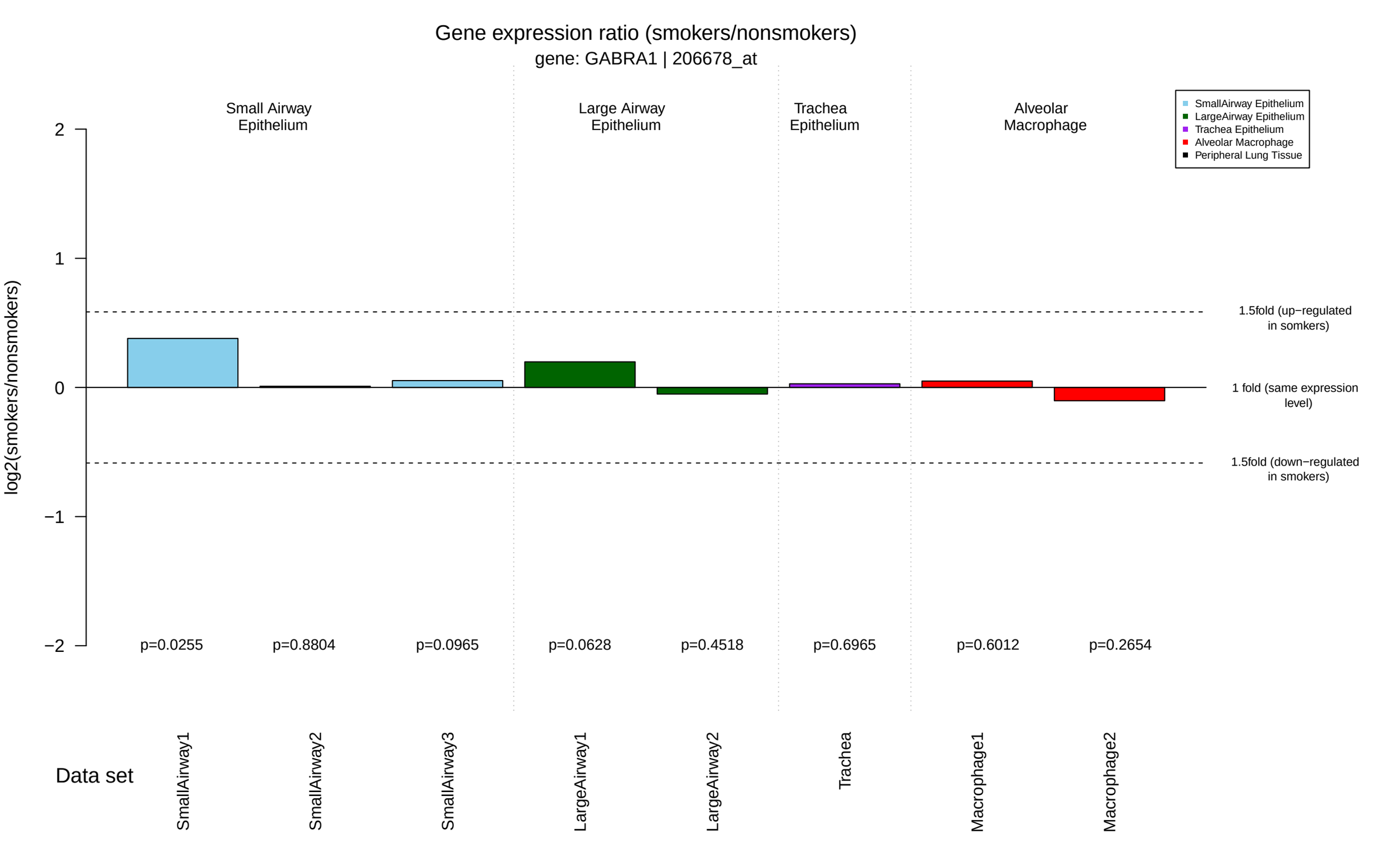
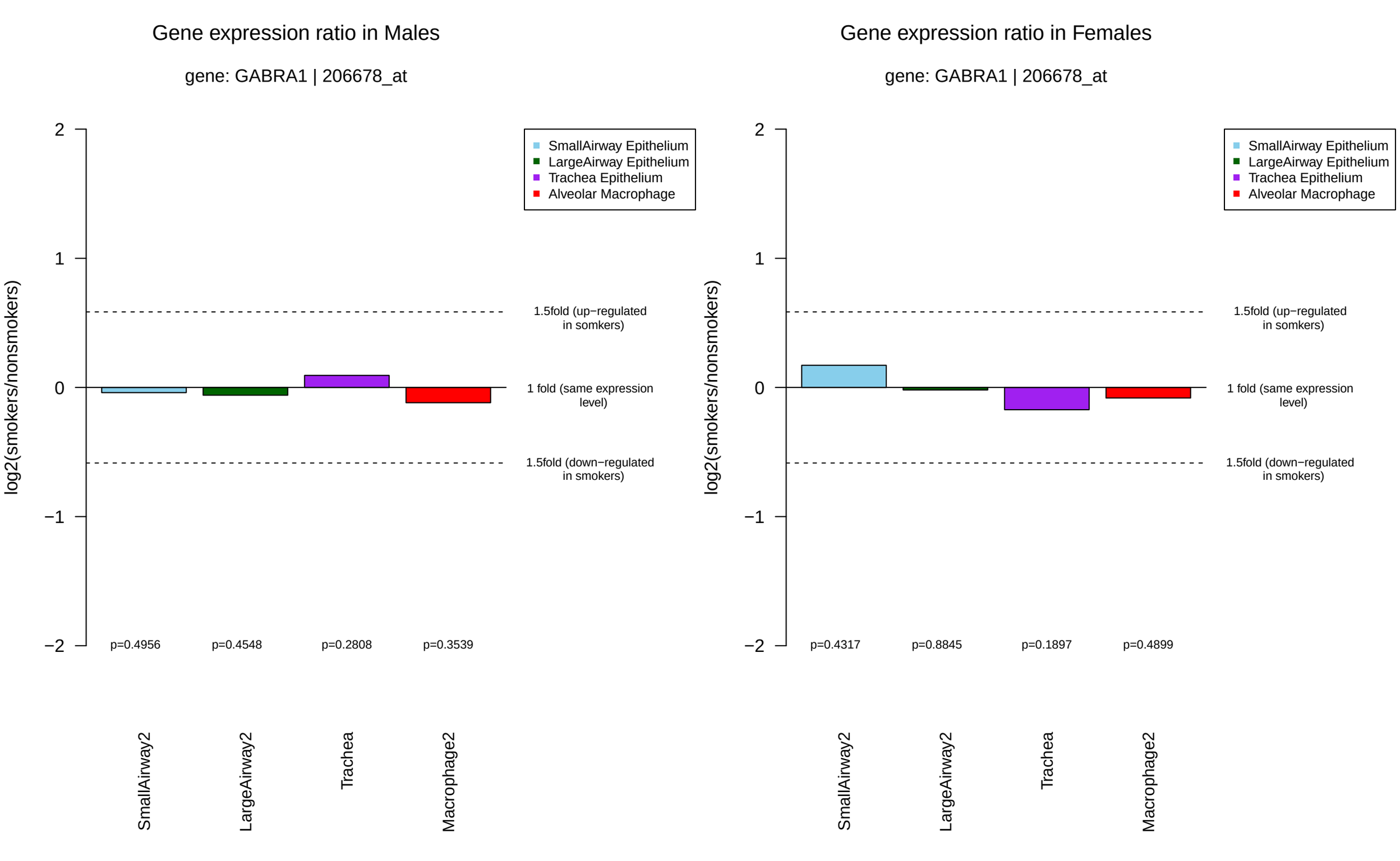
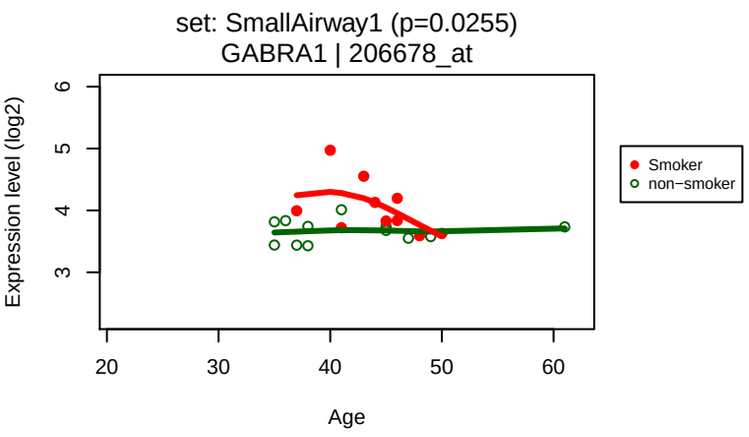
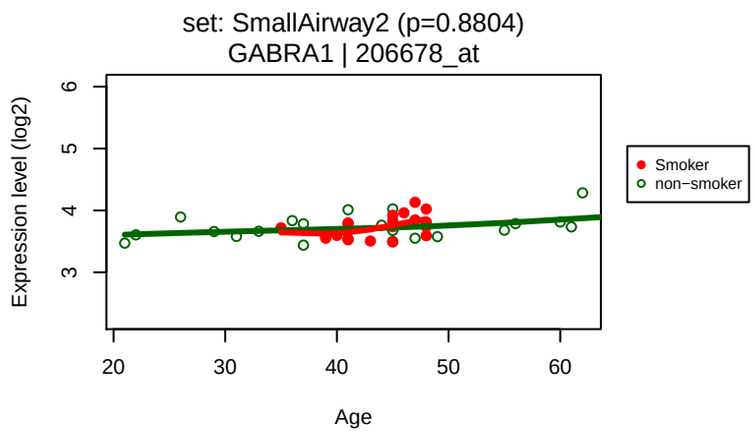
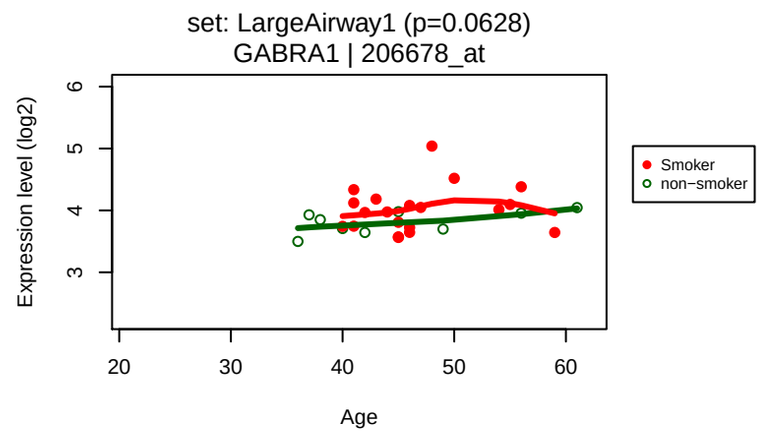

Gene expression of smokers and nonsmokers
gene: GABRA1 | 206678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0255
p=0.8804
p=0.0965
p=0.0628
p=0.4518
p=0.6965
p=0.6012
p=0.2654
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

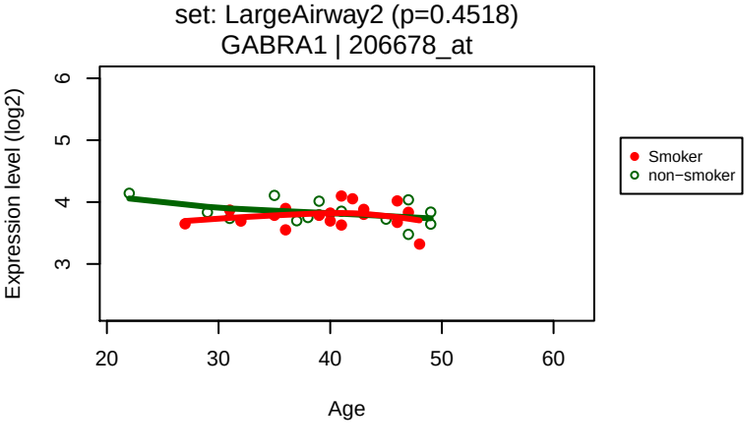

Gene expression ratio (smokers/nonsmokers)
gene: GABRA1 | 206678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0255
p=0.8804
p=0.0965
p=0.0628
p=0.4518
p=0.6965
p=0.6012
p=0.2654
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

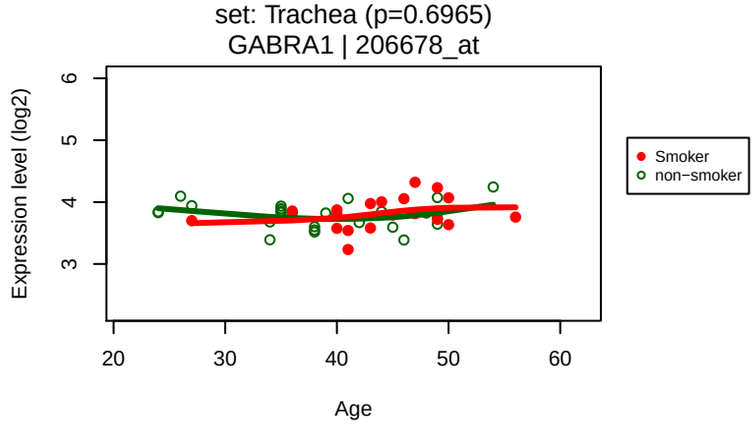

Gene expression ratio in Males
gene: GABRA1 | 206678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4956
p=0.4548
p=0.2808
p=0.3539
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRA1 | 206678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4317
p=0.8845
p=0.1897
p=0.4899
SmallAirway2
LargeAirway2
Trachea
Macrophage2

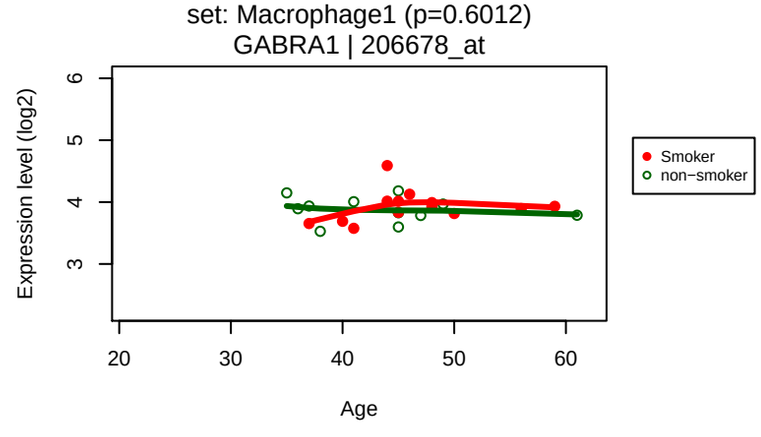

set: SmallAirway1 (p=0.0255)
GABRA1 | 206678_at
Expression level (log2)
Age
Smoker
non−smoker

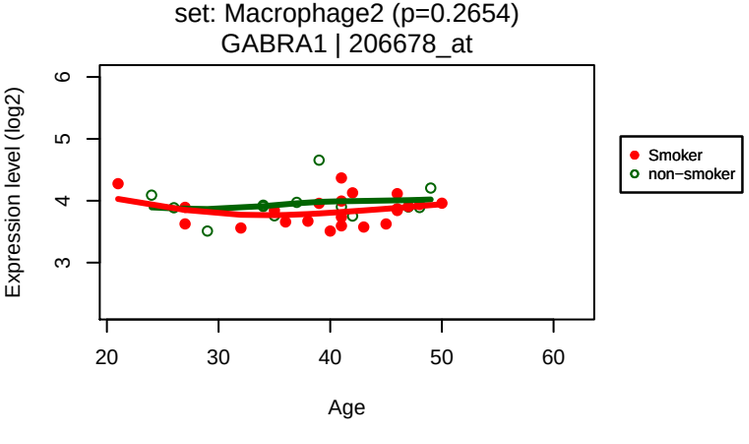

set: SmallAirway2 (p=0.8804)
GABRA1 | 206678_at
Expression level (log2)
Age
Smoker
non−smoker

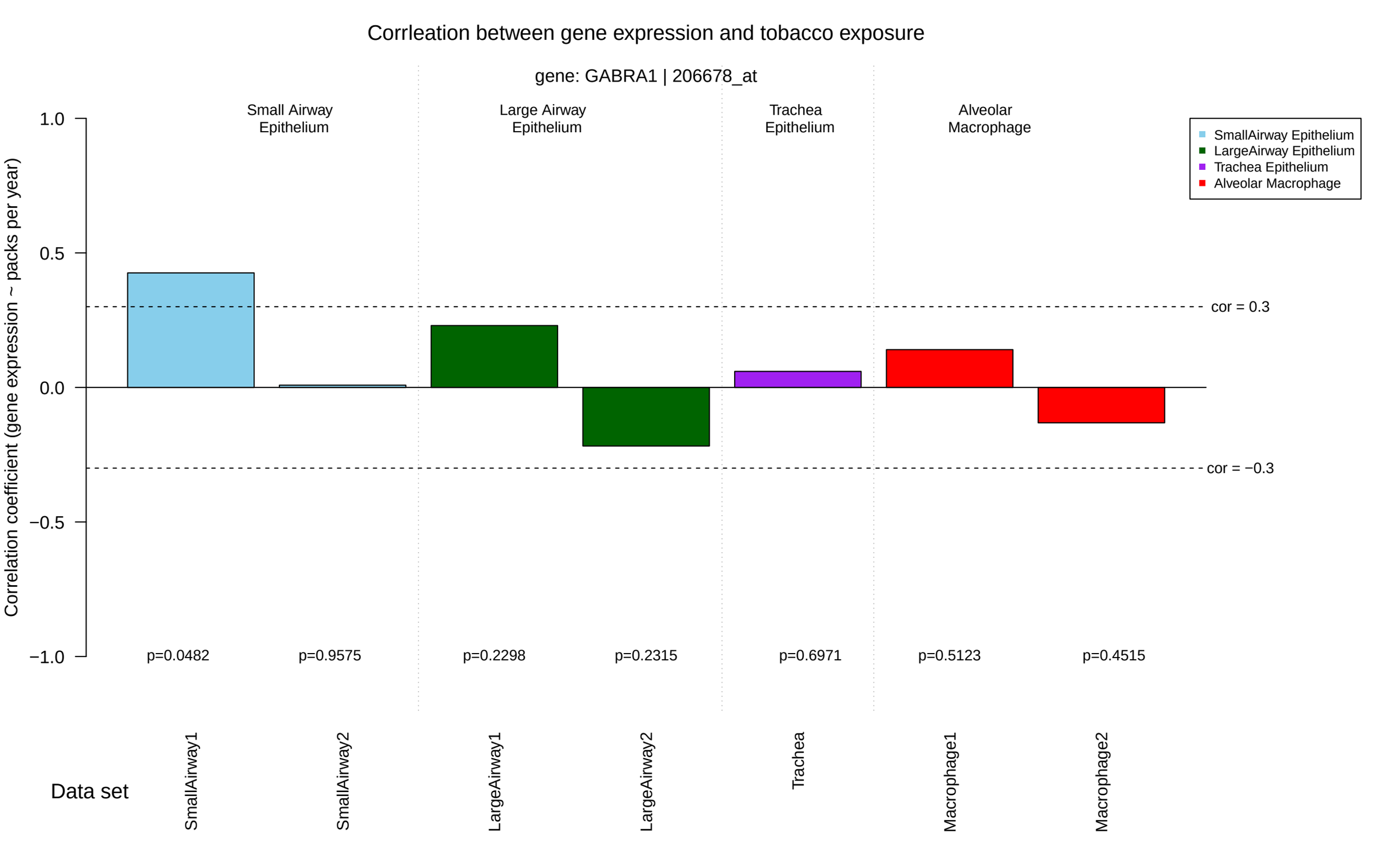

set: LargeAirway1 (p=0.0628)
GABRA1 | 206678_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4518)
GABRA1 | 206678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6965)
GABRA1 | 206678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6012)
GABRA1 | 206678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2654)
GABRA1 | 206678_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRA1 | 206678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0482
p=0.9575
p=0.2298
p=0.2315
p=0.6971
p=0.5123
p=0.4515
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2